



ECEN 314—Signals and Systems

Scott L. Miller

Module 21

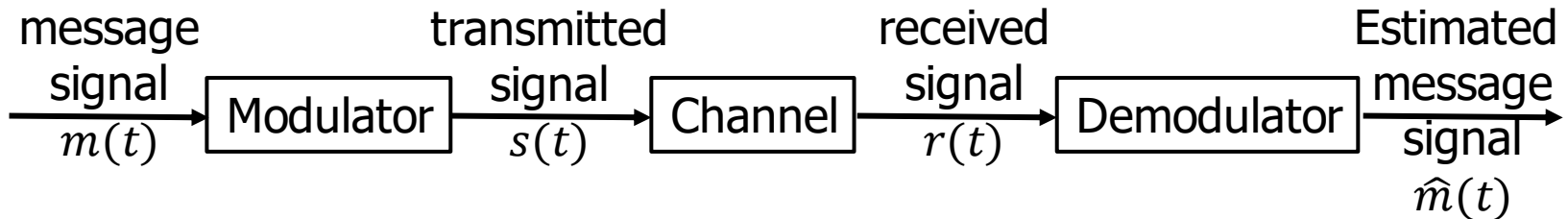
Application: Analog Communications
Part 1 - Amplitude Modulation

Learning Objectives



- Understand the basics of AM modulation. Be able to create and AM modulated signal and to demodulate an AM signal using practical circuitry.
- Understand how to create a DSB-AM signal using a double balanced modulator.
- Understand how to create an SSB-AM signal using a sideband filter.

Communication Basics



In a typical electronic communication system, we seek to convey a message from a transmitter to a receiver via some sort of communication channel.

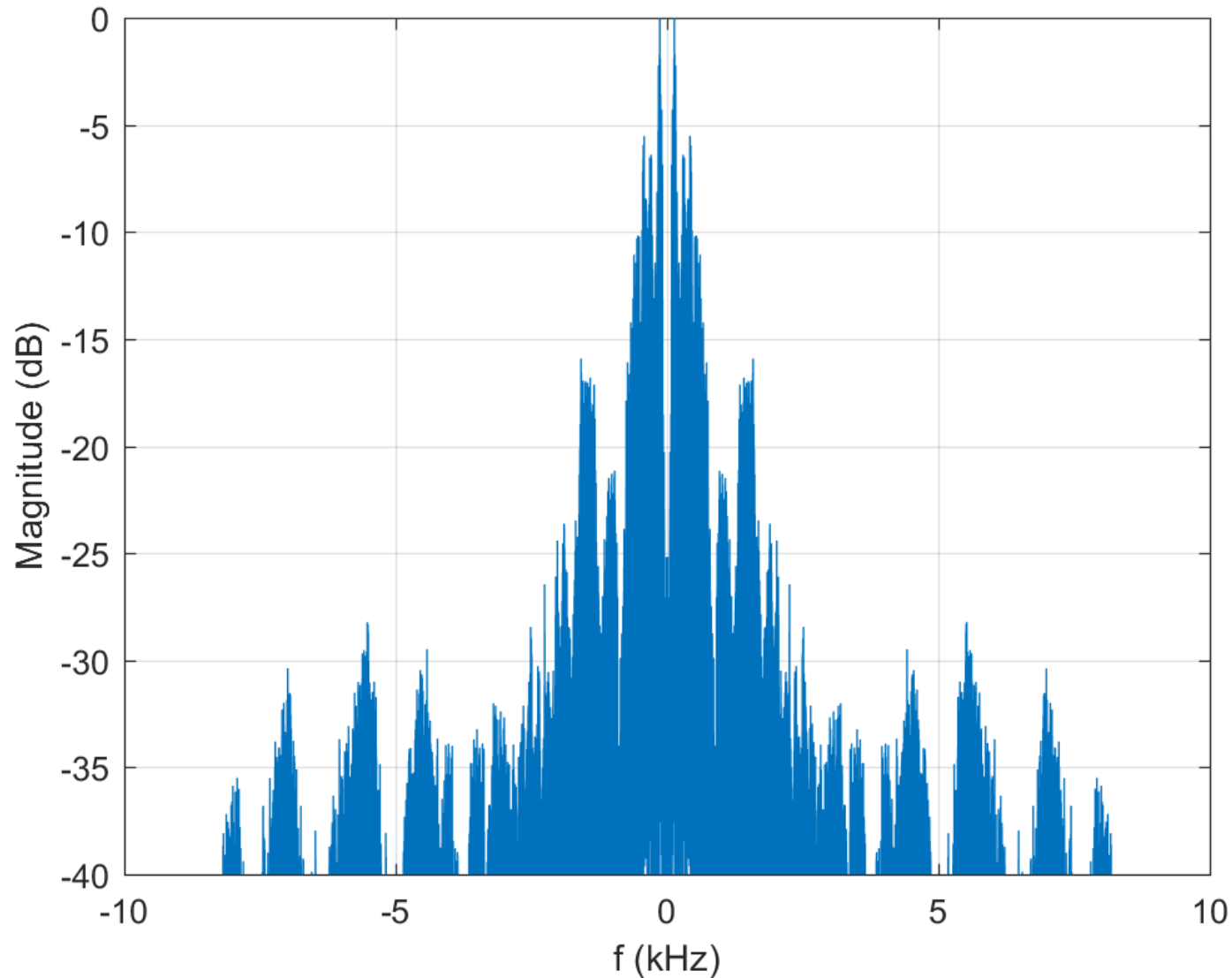
Message: A signal of some type that might be voice, music, video, etc.

Modulator: Translates the message signal into a form that is more suitable for transmission over the chosen communications medium.

Channel: The medium for getting the signal from sender to receiver. Could be twisted copper pair (landline), coaxial cable, fiber optic cable, free space (wireless), etc.

Demodulator: Extracts the desired message signal from the received signal at the receiver. Must combat the effects of noise, interference, distortion that may have occurred over the channel.

Spectrum of a Typical Voice Signal (Male)



Modulation



In wireless applications, it is necessary to translate the message to a higher frequency band as the size of the antenna needed is proportional to the wavelength used (inversely proportional to the frequency band).

Application	Frequency Band	Wavelength
AM Radio	550-1720kHz	~175-550m (575-1800 ft)
FM Radio	87.8-108MHz	~3m (10 ft)
Cellular	700, 850, 1700, 1900, 2100, 2300 MHz	~0.1-0.4m (4-16 inches)
GPS	1227.60, 1575 MHz	~0.2-0.25m (8-10 inches)
WLAN (WiFi)	2.4, 3.6, 4.9, 5, 5.9 GHz	~5-12.5cm (2-5 inches)

Amplitude Modulation (AM)



In amplitude modulation, the amplitude of a carrier (sinusoid) is proportional to the message.

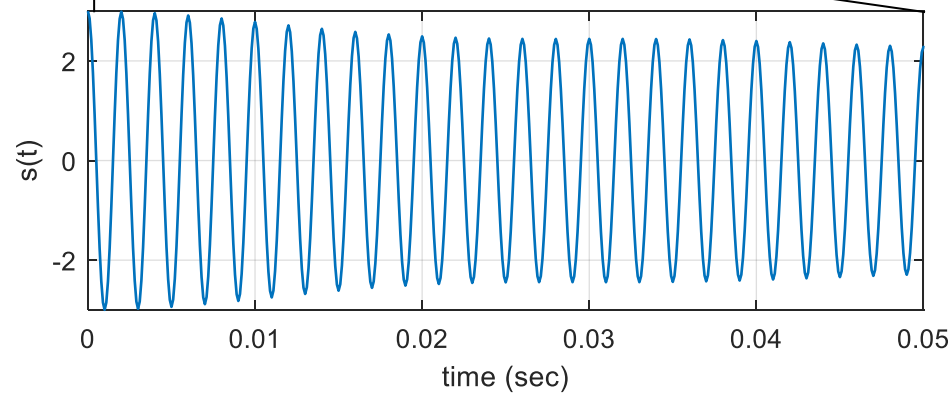
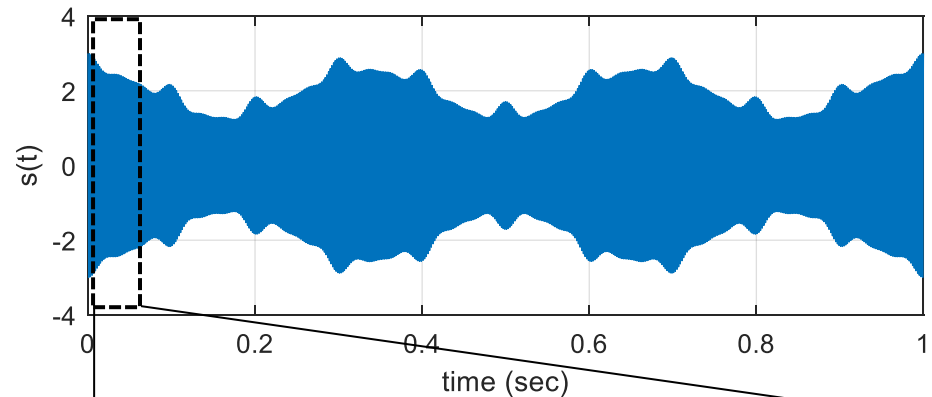
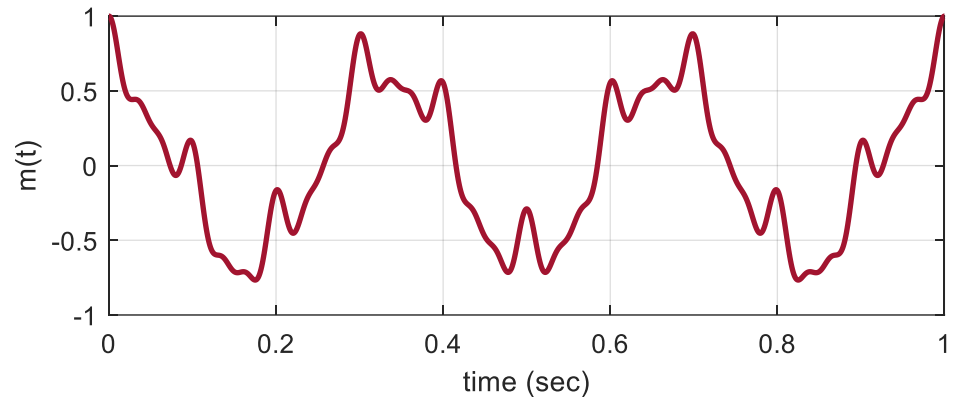
$$s(t) = A(1 + \mu \cdot m(t))\cos(\omega_c t)$$

In this expression, it is assumed that the message signal has been normalized so that its maximum value is 1 in which case the constant μ is called the *modulation index*.

The modulation index measures how much the amplitude of the AM signal varies about its unmodulated level.

In the example shown, $\mu = \frac{1}{2}$ (50%)

- Unmodulated amplitude = 2,
- Peak modulated amplitude = 3.



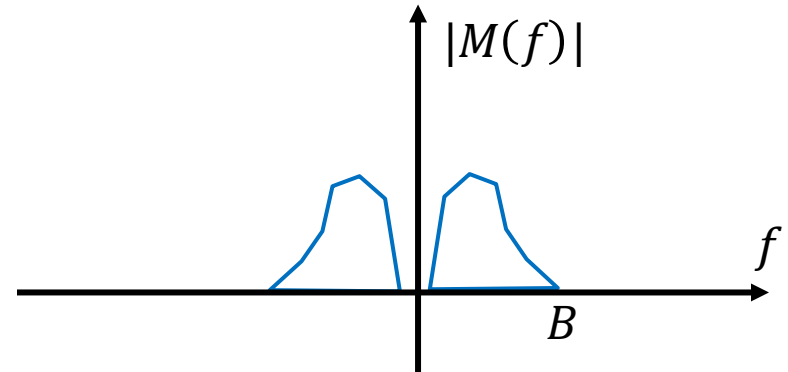
Amplitude Modulation (AM)



In the US, broadcast AM channels have:

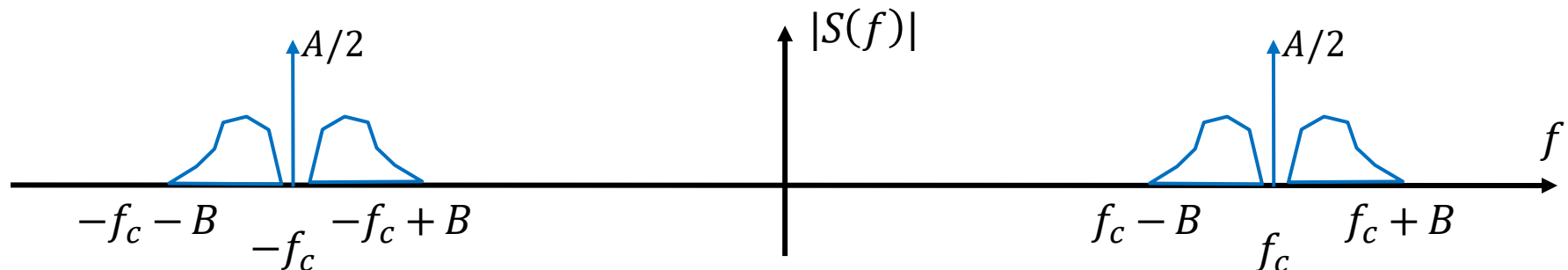
- ✓ Center frequencies = $(550 + 10m)$ kHz for $m=0, 1, 2, \dots, 117$,
- ✓ Bandwidth = 10kHz.

The bandwidth of the message signal is thus limited to no more than 5kHz.



In the frequency domain,

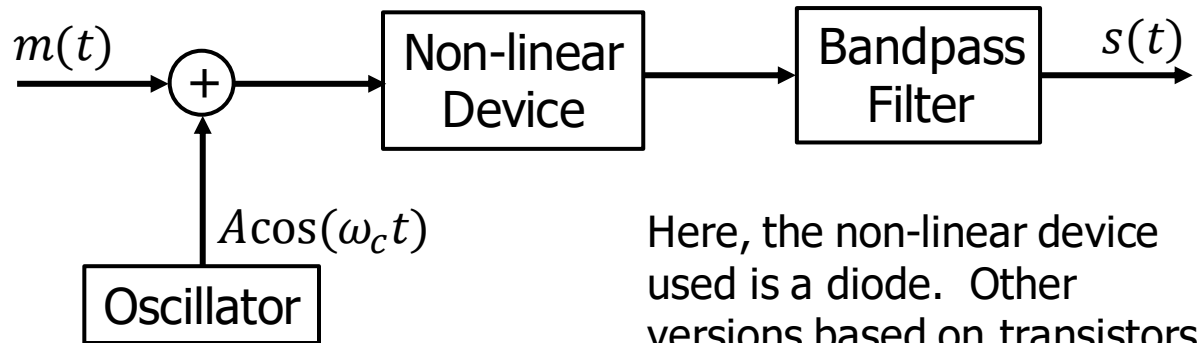
$$\begin{aligned} S(f) &= \mathcal{F}\{A(1 + \mu \cdot m(t))\cos(\omega_c t)\} \\ &= \frac{A}{2}\delta(f - f_c) + \frac{A}{2}\delta(f + f_c) + \frac{A\mu}{2}M(f - f_c) + \frac{A\mu}{2}M(f + f_c) \end{aligned}$$



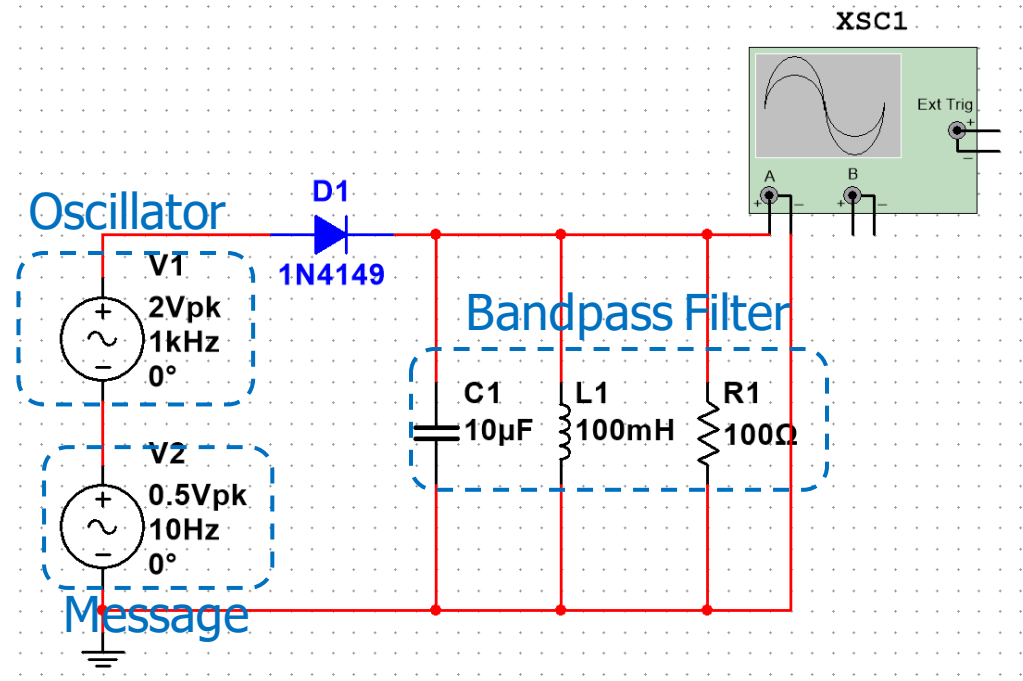
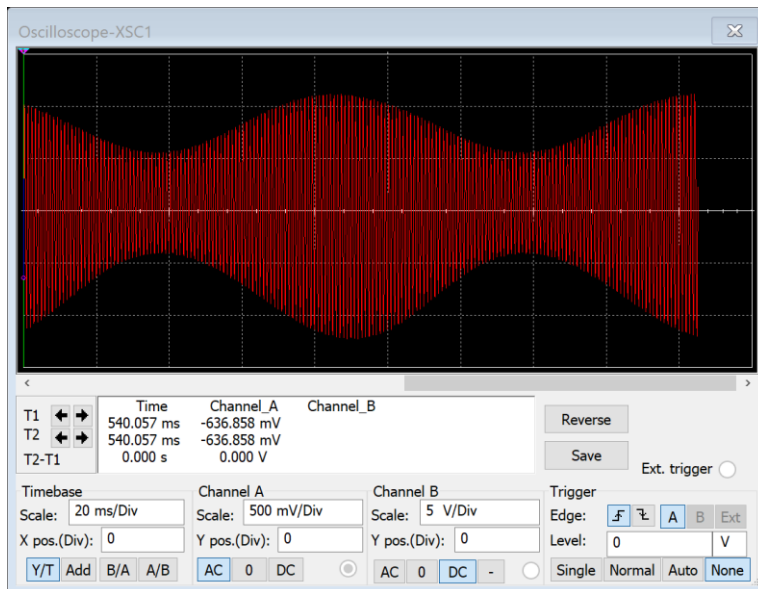
Amplitude Modulation (AM)



It is pretty simple to build a circuit to function as an amplitude modulator. These figures show one possible circuit in block diagram and schematic form.



Here, the non-linear device used is a diode. Other versions based on transistors are also possible.

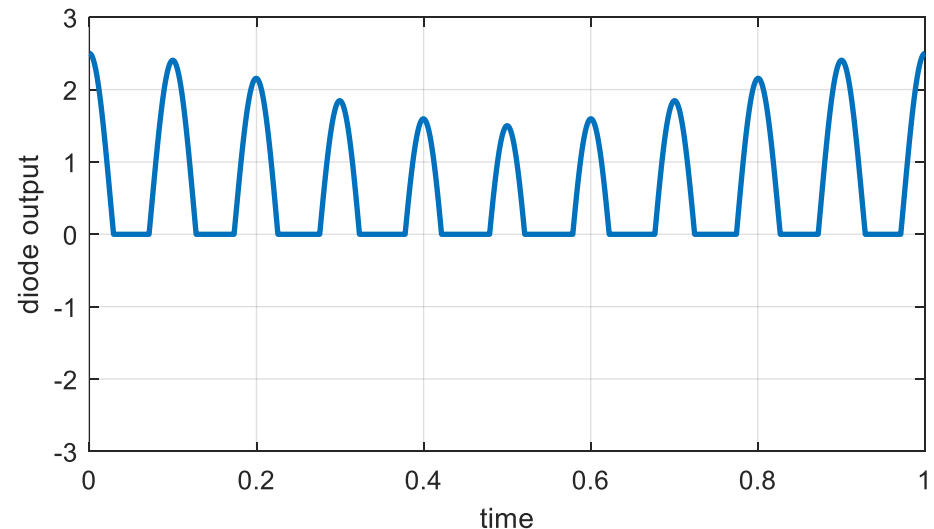
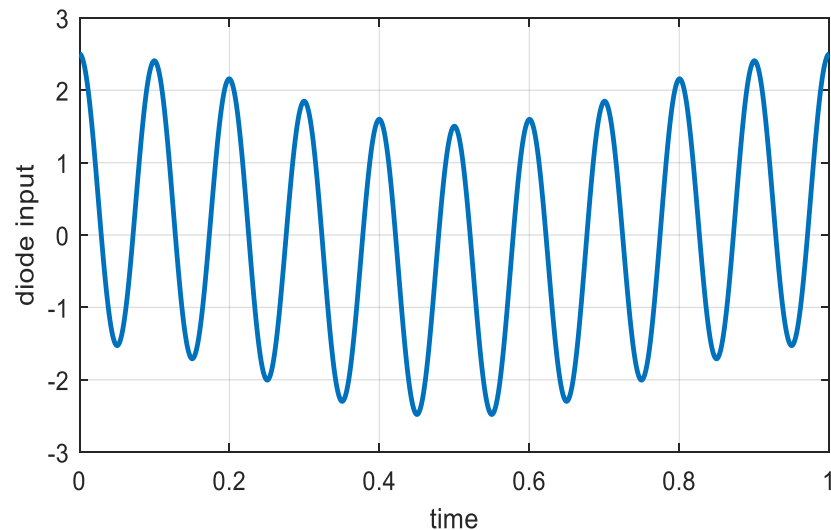
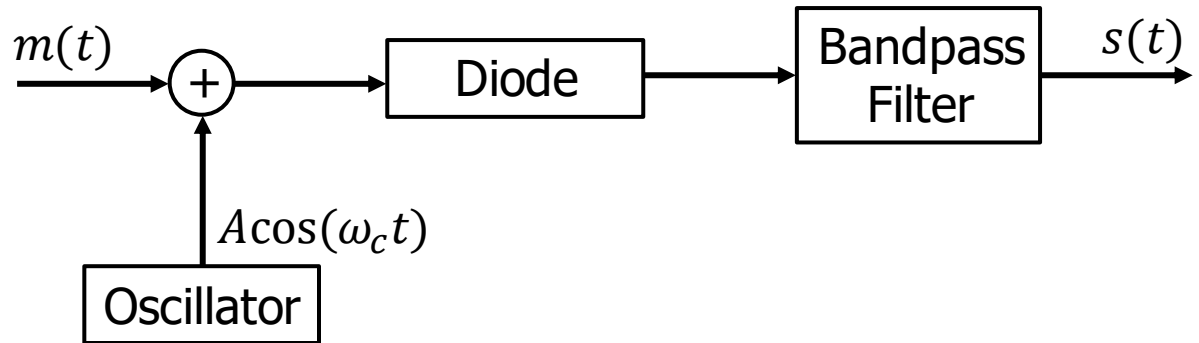


Amplitude Modulation (AM)



The diode functions as a half wave rectifier, it only passes input when the input is a positive voltage.

Since $A > m(t)$ the diode input is >0 only when $\cos(\omega_c t) > 0$.



Amplitude Modulation (AM)



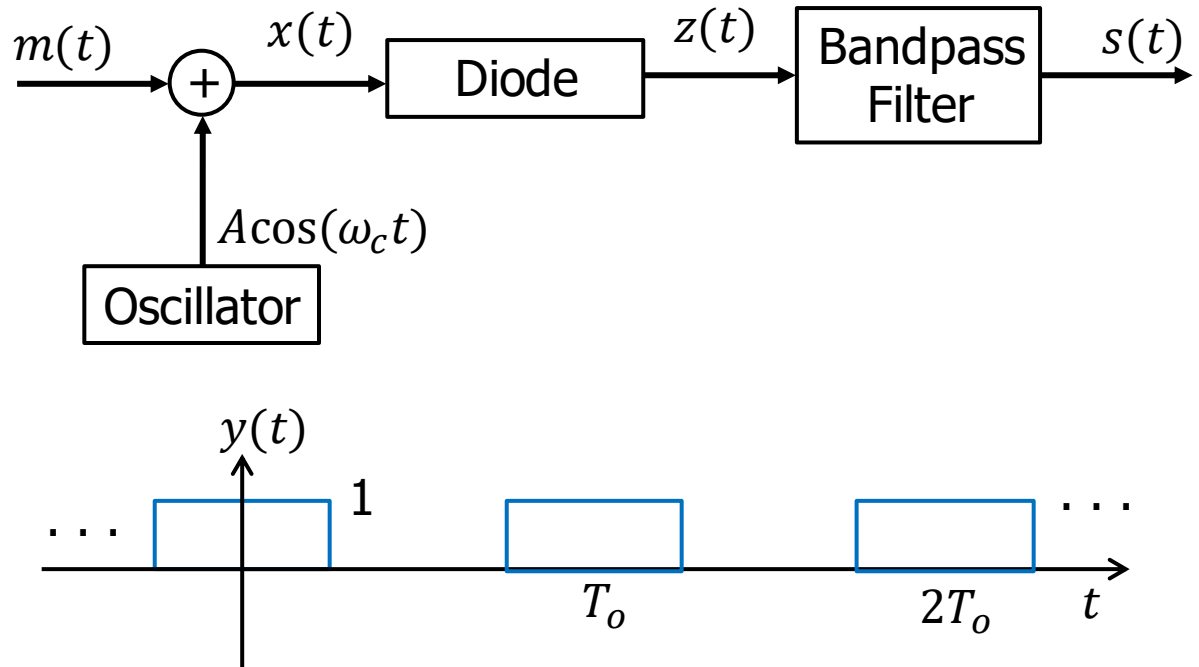
We could model the diode as multiplying its input by a periodic square wave,

$$z(t) = x(t) \cdot y(t).$$

Write $y(t)$ in terms of its F.S. representation,

$$y(t) = \sum_{k=0}^{\infty} Y_k \cos(k\omega_c t).$$

The diode output is then



$$z(t) = [A\cos(\omega_c t) + m(t)] \sum_{k=0}^{\infty} Y_k \cos(k\omega_c t)$$

$$= \underbrace{\left\{ \frac{AY_1}{2} + Y_0 m(t) \right\}}_{\text{DC part}} + \underbrace{\left\{ A \left(Y_0 + \frac{Y_2}{2} \right) + Y_1 m(t) \right\} \cos(\omega_c t)}_{\text{Terms @ or around } \omega_c} + \underbrace{\sum_{k=2}^{\infty} \left[A \left(\frac{Y_{k-1} + Y_{k+1}}{2} \right) + Y_k m(t) \right] \cos(k\omega_c t)}_{\text{Terms @ or around } k\omega_c}$$

DC part

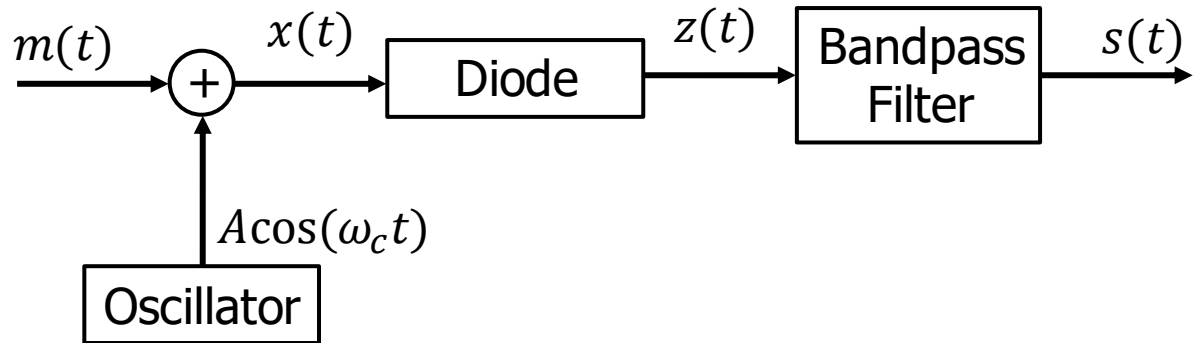
Terms @ or around ω_c

Terms @ or around $k\omega_c$

Amplitude Modulation (AM)



The BPF then selects only the terms at or near ω_c and blocks all the other terms. The resulting output of the BPF is an AM signal.

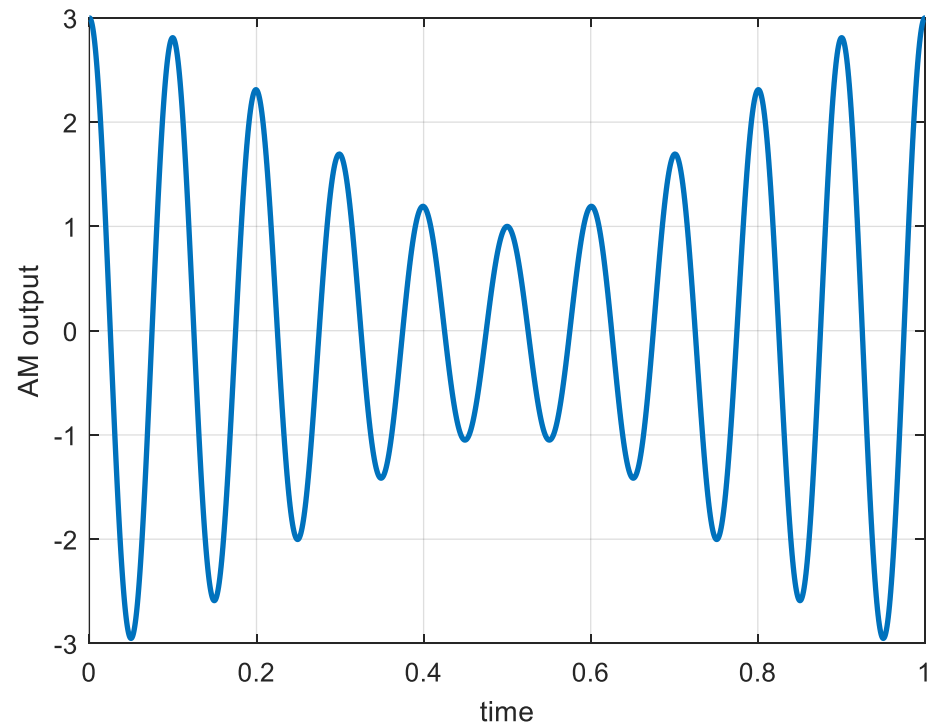


$$s(t) = \left\{ A \left(Y_o + \frac{Y_2}{2} \right) + Y_1 m(t) \right\} \cos(\omega_c t)$$

The modulation index of this AM signal is

$$\mu = \frac{Y_1}{A \left(Y_o + \frac{Y_2}{2} \right)}.$$

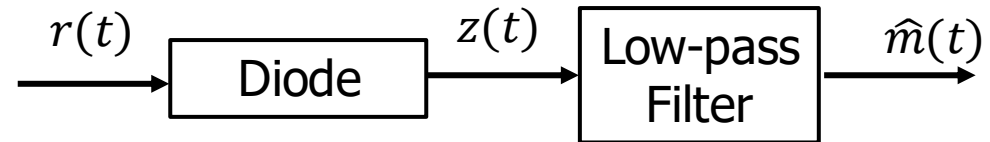
Since the F.S. coefficients are all fixed and known, we can produce any modulation index we want by choosing an appropriate value of A .



AM Demodulation



At the receiving end, AM demodulation works much like the modulation process, except this time we select the DC part with a low-pass filter.



$$r(t) = A\{1 + \mu m(t)\}\cos(\omega_c t)$$

$$\begin{aligned} z(t) = r(t)y(t) &= A\{1 + \mu m(t)\}\cos(\omega_c t) \left[Y_0 + \sum_{k=1}^{\infty} Y_k \cos(k\omega_c t) \right] \\ &= \frac{AY_1}{2} \{1 + \mu m(t)\} + \text{terms @ } \omega_c \text{ and higher} \end{aligned}$$

The LPF removes the high frequency terms and only the DC parts remain so that

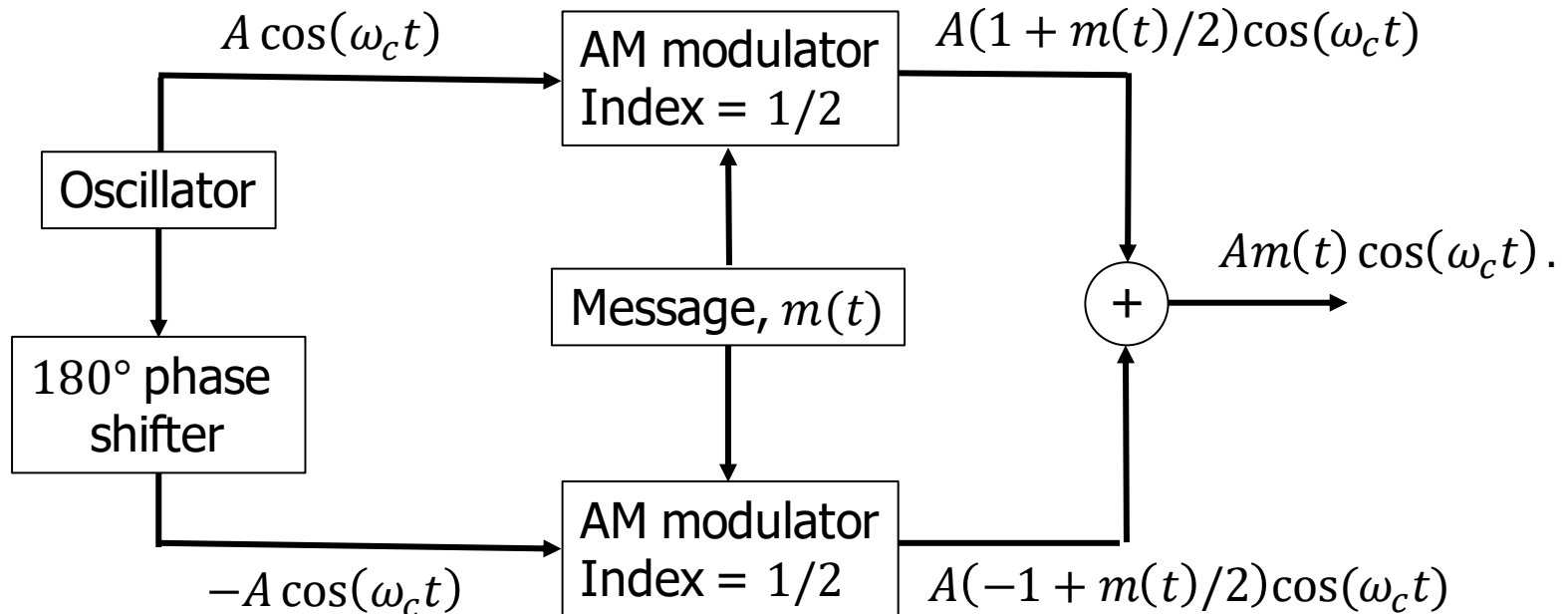
$$\hat{m}(t) = \frac{AY_1}{2} \{1 + \mu m(t)\}.$$

As long as the message has no DC component (which is typically the case for voice, music, etc.), we can block the DC part (with a capacitor) and end up with $\hat{m}(t) \propto m(t)$.

Another form of AM modulation is known as double side-band (suppressed carrier) modulation. In this case, the modulated signal is

$$s(t) = Am(t) \cos(\omega_c t).$$

In DSB there is no separate carrier term, just the term with the message. One could construct a DSB modulator using two balanced AM modulators.

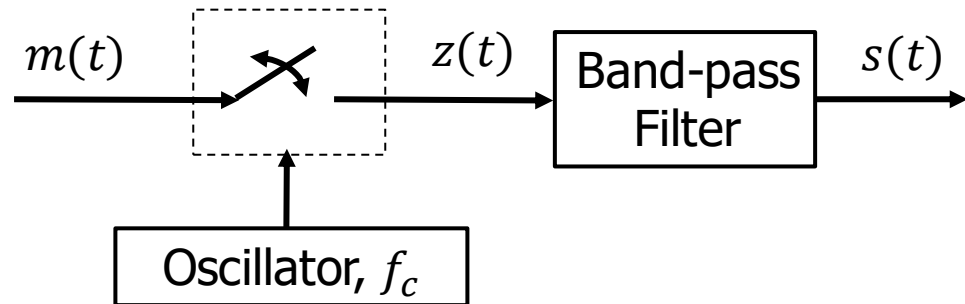


DSB-sc AM



Another way to create a DSB signal is using what is known as a switching modulator.

The oscillator controls the operation of the switch which opens and closes at a frequency of f_c Hz (ω_c rad/sec).

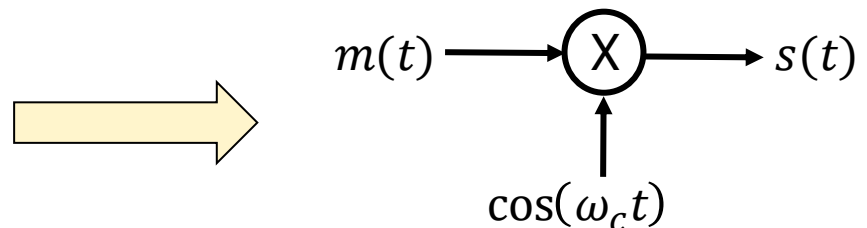


As before, we write $z(t) = m(t)y(t)$ where $y(t)$ is a periodic square wave. Then

$$z(t) = m(t) \sum_{k=0}^{\infty} Y_k \cos(k\omega_c t).$$

The BPF then selects the term at frequency ω_c and blocks the others producing $s(t) \propto m(t) \cos(\omega_c t)$.

Regardless of how we actually build the DSB modulator, in the future we will represent such a modulator as a multiplier in our block diagrams.



DSB-sc AM



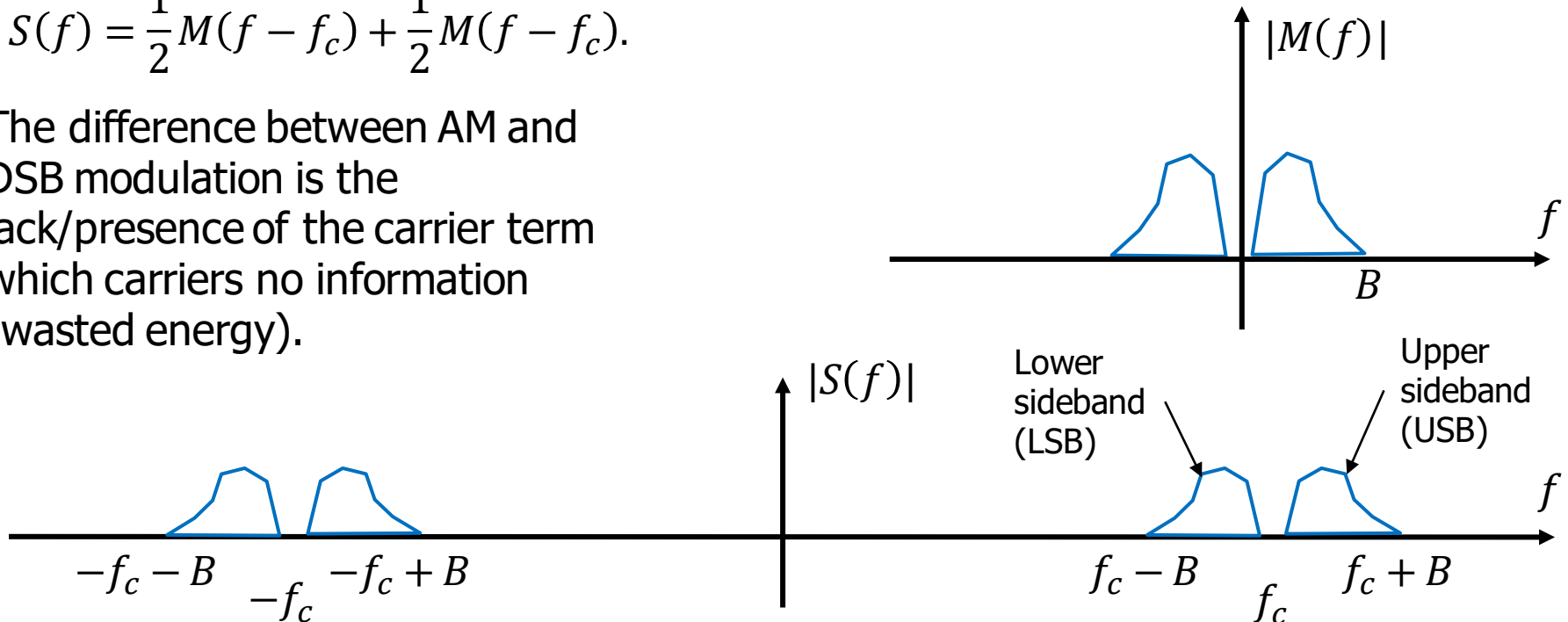
In the frequency domain, the DSB modulator just shifts the spectrum of the message signal to the desired frequency band.

$$s(t) = m(t) \cos(\omega_c t),$$

$$S(f) = \frac{1}{2}M(f - f_c) + \frac{1}{2}M(f + f_c).$$

The difference between AM and DSB modulation is the lack/presence of the carrier term which carries no information (wasted energy).

Note: Both AM and DSB modulation requires a channel bandwidth of $2B$ to transmit a message whose bandwidth is B .



DSB Demodulation



Because the DSB signal lacks the discrete carrier component of AM, it is a bit more difficult to demodulate.

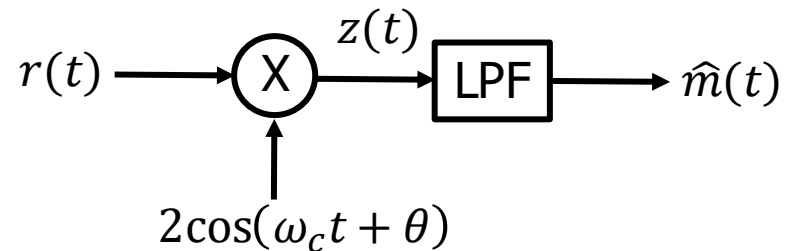
Assuming the received signal is a DSB signal:

$$r(t) = m(t) \cos(\omega_c t),$$

$$\begin{aligned} z(t) &= 2m(t) \cos(\omega_c t) \cos(\omega_c t + \theta) \\ &= m(t) \cos(\theta) + m(t) \cos(2\omega_c t + \theta). \end{aligned}$$

The LPF selects the term at and around DC so that

$$\hat{m}(t) = m(t) \cos(\theta)$$



Note: The oscillator used in the demodulator must be synchronous in phase ($\theta \approx 0$) with the oscillator used in the modulator.

Since the transmitter and receiver are not at the same location, achieving phase synchronization can be tricky. A device known as a phase lock loop is used to achieve this purpose.

SSB Modulation



Assuming the messages we send are real signals, then their spectra are symmetric

$$M(f) = M^*(-f)$$

- Even amplitude symmetry,
- Odd phase symmetry.
- ✓ There is no information in the LSB that is not in the USB (and vice versa).
- ✓ It is not really necessary to transmit both sidebands.
- ✓ We could save bandwidth by sending only one of the sidebands → single sideband (SSB) modulation.
- ✓ Implementing the ideal (or nearly ideal) sideband filter may be impractical. Alternative methods are used.

